

第零讲 人工智能历史

Lecture 0 History of Artificial Intelligence

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智能

Intelligence

- ▶ 关于“智能”一词的起源，它源自拉丁词“*intelligentia*”（或*intellegentia*），由“*inter*”（在.....之间）和“*legere*”（选择、辨别或阅读）构成。该词最初传达了在不同事物之间进行选择、区分或理解的含义。随着时间的推移，其内涵逐渐扩展，涵盖了认知能力、理解能力、学习能力、推理能力、问题解决能力以及环境适应能力等多个方面。在当前的讨论中，“智能”通常指一个实体感知信息、将其保留为知识，并以有目的的方式加以运用，从而适应环境的能力。

Regarding the origin of the term “intelligence,” it derives from the Latin word *intelligentia* (or *intellegentia*), which is composed of *inter* (“between” or “among”) and *legere* (“to choose,” “discern,” or “read”). The term originally conveyed the idea of selecting, distinguishing, or understanding among different things. Over time, its meaning evolved to encompass various aspects of cognition, including understanding, learning, reasoning, problem-solving, and adaptability. In the present discussion, intelligence generally refers to the ability of an entity to perceive information, retain it as knowledge, and purposefully apply it in order to adapt to its environment.

智能

Intelligence

- 历史上曾有不少学者对智能的定义与内涵给出过相似或不同的理解，相关讨论在古今中外都不曾停息：

古希腊哲学：

- **柏拉图：**柏拉图认为智能是最高形式的知识，涉及理解超越物质世界的永恒真理和现实。他相信灵魂在出生前拥有内在的知识。
- **亚里士多德：**亚里士多德将智能视为理性思维和逻辑推理的能力。他强调通过经验观察和实践学习来发展智能的重要性。

- Throughout history, numerous scholars have offered similar or contrasting definitions and insights into the nature of intelligence, sparking a continuous debate that spans from ancient times to the modern era, and across both Eastern and Western cultures:

Ancient Greek Philosophy:

- **Plato:** Plato regarded intelligence as the highest form of knowledge, involving the comprehension of eternal truths and realities that transcend the material world. He believed that the soul possesses innate, intrinsic knowledge acquired prior to birth.
- **Aristotle:** Aristotle viewed intelligence as the capacity for rational thought and logical reasoning. He emphasized the fundamental importance of developing intelligence through empirical observation and practical, experiential learning.

智能 Intelligence

中世纪学院哲学：

- **托马斯·阿奎那：**在中世纪，托马斯·阿奎那将亚里士多德的哲学与基督教神学融合在一起。他将智能定义为理解真理和参与沉思的能力，特别涉及神圣事务。

文艺复兴人文主义：

- **文艺复兴思想家：**在文艺复兴时期，如彼特拉克和伊拉斯谟，强调全面教育的重要性以及多种才能的发展作为智能的标志。他们相信批判性思维、语言技能和艺术能力的价值。

Medieval Scholastic Philosophy:

- **Thomas Aquinas:** During the Middle Ages, Thomas Aquinas synthesized Aristotelian philosophy with Christian theology. He defined intelligence as the capacity to comprehend truth and engage in deep contemplation, with a particular focus on the divine realm.。

Renaissance Humanism:

- **Renaissance Thinkers:** Thinkers of the Renaissance period, such as Petrarch and Erasmus, emphasized the importance of a well-rounded education and viewed the cultivation of diverse talents as the hallmark of intelligence. They placed a high value on critical thinking, linguistic skills, and artistic capabilities.

智能

Intelligence

启蒙时代理性主义：

- ▶ 约翰·洛克：洛克的经验主义哲学认为智能是感官经验的结果，以及通过观察和反思积累知识的产物。
- ▶ 让-雅克·卢梭：卢梭与洛克相反，认为真正的智能在于接纳自然的倾向和情感，而不是符合社会规范。

19世纪心理学：

- ▶ 弗朗西斯·高尔顿：在19世纪，高尔顿专注于智能测量。他将智能定义为感官敏锐度、反应时间和认知能力。他引入了优生学概念，认为智能主要是遗传的。

20世纪初：

- ▶ 阿尔弗雷德·比奈：比奈在智力测试方面的工作导致了第一个智商测试的发展。他将智能定义为推理、解决问题和适应新情况的能力。
- ▶ 刘易斯·特曼：特曼改编了比奈的智商测试，并推广了智商指数（IQ）的概念。他认为智能是一种与生俱来、稳定的特质。

智能 Intelligence

Enlightenment Philosophy:

- ▶ **John Locke:** Locke's empiricist philosophy posited that intelligence is a product of sensory experience, asserting that knowledge is accumulated entirely through observation and reflection.
- ▶ **Jean-Jacques Rousseau:** In contrast to Locke, Rousseau argued that true intelligence lies in embracing one's natural inclinations and emotions, rather than conforming to rigid societal norms.

19th-Century Psychology:

- ▶ **Francis Galton:** In the nineteenth century, Galton focused heavily on the measurement of intelligence, defining it through sensory acuity, reaction time, and cognitive capacity. He introduced the concept of eugenics, maintaining that intelligence is predominantly hereditary.

Early 20th Century:

- ▶ **Alfred Binet:** Binet's pioneering work in mental testing led to the development of the first practical intelligence test. He defined intelligence as the capacity to reason, solve problems, and adapt to novel situations.
- ▶ **Lewis Terman:** Terman adapted Binet's work to create the Stanford-Binet test and popularized the concept of the Intelligence Quotient (IQ). He conceptualized intelligence as an innate, stable, and immutable trait.

智能

Intelligence

- 历史上，人们多次提出创造人工实体或机制来代替人类执行任务的想法。尽管这些实例可能并未直接导致我们今天所理解的人工智能的发展，但它们确实反映出人类渴望创造能够模仿某些智能或劳动能力的工具和装置：

古代的自动机： 在古代文明中，如埃及、希腊和中国，有关于能够执行特定任务的机械装置和自动机的记载。这些装置通常通过简单的机械原理驱动，虽然并不具备现代意义上的智能，但它们展示了早期尝试创造自动化系统的努力。

炼金术和早期工程： 在中世纪和文艺复兴时期，炼金术士和工程师致力于创造能够模仿某些过程的装置。虽然与现代人工智能无直接关系，但这些努力为科学探索和实验奠定了基础。

雅克·德·沃克松的自动机： 在18世纪，法国发明家雅克·德·沃克松创造了复杂的机械自动机，其中包括一个能够进食、拍动翅膀和消化食物的机械鸭。尽管这些创造并不是现代意义上的智能，但它们展示了机械装置模仿某些行为的潜力。

智能 Intelligence

- ▶ Throughout history, humanity has repeatedly conceived ideas of creating artificial entities or mechanisms to perform tasks in place of human beings. Although these early instances did not directly cause the development of artificial intelligence as we understand it today, they reflect an enduring human desire to forge tools and devices capable of mimicking certain aspects of intelligence or physical labor:

Ancient Automata: Ancient civilizations, including Egypt, Greece, and China, contain historical records of mechanical devices and automata designed to execute specific tasks. Driven primarily by basic mechanical principles, these apparatuses possessed no intelligence in the modern sense; nevertheless, they represent humanity's earliest endeavors to build automated systems.

Alchemy and Early Engineering: During the Middle Ages and the Renaissance, alchemists and engineers dedicated themselves to crafting devices that could simulate biological or natural processes. While not directly related to modern AI, these persistent efforts laid a critical foundation for scientific exploration and empirical experimentation.

Jacques de Vaucanson's Automata: In the 18th century, the French inventor Jacques de Vaucanson constructed extraordinarily complex mechanical automata, most famously a mechanical duck capable of eating, flapping its wings, and simulating digestion. Although these creations were by no means intelligent by modern standards, they powerfully demonstrated the potential of machinery to imitate lifelike behaviours.

智能 Intelligence

工业革命： 工业革命的到来导致了各种能够提高生产力并减少某些人工劳动需求的机器和工具的发展。虽然并非专注于创造智能机器，但这个时代标志着对待工作的方法和效率发生重大变革。

查尔斯·巴贝奇和艾达·洛芬： 在19世纪，查尔斯·巴贝奇设计了"分析引擎"，一种机械计算机的前身。他的合作者艾达·洛芬编写了被认为是第一个旨在由机器处理的算法，展示了对机器能够进行复杂计算的潜在认识。

The Industrial Revolution: The advent of the Industrial Revolution led to the development of various machines and tools capable of boosting productivity and reducing the demand for certain forms of manual labour. Although not focused on creating intelligent machinery, this era marked a profound transformation in societal approaches to labour and operational efficiency.

Charles Babbage and Ada Lovelace: In the 19th century, Charles Babbage designed the "Analytical Engine," a mechanical precursor to the modern computer. His collaborator, Ada Lovelace, wrote what is widely considered the first algorithm intended to be processed by a machine, demonstrating a visionary understanding of a machine's potential to execute complex computations.

起源

Origination

- ▶ 尽管历史中对智能的概念的实现有过探讨和尝试，但直到20世纪50年代，人工智能的概念才正式出现，逐渐登上学术与业界的舞台。这和同期科技、思想和社会发展是密切相关的。我们从以下学科的发展来探讨人工智能建立的条件。注意这样的讨论并不能涵盖所有的因素，但确实能在一定程度上给人工智能的出现与发展提供一定的洞见。

- ▶ 数学
- ▶ 生理学
- ▶ 控制论
- ▶ 其它方面

- ▶ Although various historical discussions and attempts were made to realize the concept of intelligence, it was not until the 1950s that artificial intelligence was formalized and introduced to the scholarly and engineering arenas. Driven by the technological insights and breakthroughs of that era, the emergence of AI was shaped by several critical factors. The following sections will explore the foundational conditions necessary for the establishment of artificial intelligence. While this discussion is not exhaustive, it provides substantial insight into the emergence and early evolution of the field.

- ▶ Mathematics
- ▶ Physiology
- ▶ Cybernetics
- ▶ Miscellaneous

数学

Mathematics

- ▶ 由于数学的精确性、清晰性与严谨性，人们向来由将概念、事务、过程进行数学化的倾向，包括智能。这个方面的努力的代表为逻辑学的建立，以及包括可解性或可计算性的研究：

逻辑

- ▶ **乔治·布尔**：用于进行逻辑推理的形式语言
- ▶ **戈特洛布·弗雷格**：一阶逻辑 (FOL)

可计算理论

- ▶ **库尔特·哥德尔**：判定性理论
- ▶ **艾伦·图灵**：图灵机与可计算理论
- ▶ **阿隆佐·丘奇**：图灵完备性理论

- ▶ Due to the precision, clarity, and rigor of mathematics, humanity has always possessed an inherent inclination to mathematize concepts, entities, and processes—and intelligence is no exception. Prominent milestones in this endeavor include the establishment of mathematical logic and foundational research into decidability and computability:

Logic

- ▶ **George Boole**: Developed a formal language for logical reasoning (Boolean algebra).
- ▶ **Gottlob Frege**: Pioneered modern mathematical logic and formulated First-Order Logic (FOL).

Computability Theory

- ▶ **Kurt Gödel**: Formulated the Incompleteness Theorems.
- ▶ **Alan Turing**: Turing machine model and the foundational concepts of computability.
- ▶ **Alonzo Church**: Developed the λ -calculus and co-authored the Church-Turing Thesis

数学

Mathematics

- ▶ 除此之外，数学的多个分支，如代数、分析、几何、微分方程、概率统计等多个方面，在上世纪初便蓬勃发展，更多地从公理化、结构、表示等方便研究问题，到上世纪40~50年代，人们已基本建立了现代数学的基石。
- ▶ 虽然人们由从数学上进行建模解决问题的倾向，但对如下问题，当时包括现代的数学不能说无能为力，但至少是捉襟见肘的：
 - ▶ In addition, other branches of mathematics—such as mathematical analysis, geometry, differential equations, probability, and statistics—experienced rapid and prosperous development over the last century. From the axiomatic structuring of these fields to the deep exploration of complex foundational problems, these cumulative advancements ensured that the fundamentals of modern mathematics were firmly established by the 1940s and 1950s.
 - ▶ While humanity has an inherent inclination to use mathematical modeling to solve problems, certain tasks proved exceptionally resilient to traditional approaches. It was not that these problems were completely intractable, but rather that conventional mathematical efforts yielded highly limited results

感知（Perception）		自然语言（Natural Language）			推理（Reasoning）
视觉 （Vision）	语音 （Speech）	理解 （Understanding）	生成 （Generation）	翻译 （Translatation）	

数学

Mathematics

- ▶ 这种基于逻辑或其它数学理论的建模方式，不太成功的原因有如下几点：
 - ▶ 对于上述这种对人类而言的平凡任务，实际上可能是高度非线性的。我们并不清楚如何建立一个数学模型来描述，甚至不太确定这样的模型到底存在不存在，即模型的存在性问题；
 - ▶ 假设这个模型存在，我们可能不确定怎样对这个模型进行求解。根据人们研究非线性问题的经验，模型就算有解，在当时的算力情况下可能也无法解出，即模型的可解性问题；
 - ▶ 假设这个模型存在且可解，我们并不确信一个模型的结果能否平滑地迁移到与之类似的问题上。假设可以迁移，模型对于新问题还在多大程度上有效，即模型的扩展性问题。
- ▶ 上述这些平凡任务对我们人类来说是轻而易举的，因此，审视大脑的工作原理从而建立与之类似的处理模型来解决这些问题，这个想法与动机是自然而然的。

数学

Mathematics

- ▶ This modelling method based on logic or other mathematical theories is not that successful due to the following reasons:
 - ▶ First these mundane tasks for human beings actually are highly nonlinear and we may have no idea how to set up mathematical models to describe them. We are even not sure whether such models exist or not, aka., existence of models.
 - ▶ Second suppose there exist corresponding models, but according to people's experience in solving nonlinear problems, even the model have some solutions, due to the computing capacity back then, people could not solve them, aka., the solubility of models.
 - ▶ Third, suppose we could model the problems and obtain solutions, but we could not guarantee that the solution of one model can migrate smoothly to another similar problem and still retain the efficacy, aka., generalization of models.
- ▶ However, these mundane tasks seems trivial for our human beings. Therefore, to understand the working principles of our brains to inspire similar models to process these problems, such idea was very natural and straightforward.

生理学

Physiology

- ▶ 生理学在发展的早期，与哲学、心理学、医学并没有特别清晰的界限，其更像一种为这些学科服务的手段。例如，借助对人体的了解，构建一个机械设备来执行心智操作，以验证机械唯物主义；了解大脑如何存储与处理信息，加深对认知心理学的研究；解剖与研究人体结构，为了更好地治病救人。
- ▶ 但科技的发展，实验手段的丰富，还是促成了生理学研究的深入，例如，在神经生理学方面，借助于染色与显微技术，对中枢与周边神经系统进行了深入的研究，确立了大脑是意识基础的中心地位；意识到简单的神经元群体行为可以引发思维和行动。
- ▶ 除此之外，一些学者如赫尔曼·冯·亥姆霍兹（1821-1894），威廉·冯特（1832-1920），也采用了一些科学的方法对人的视觉系统进行了研究。
- ▶ 到上世纪40年代，基本建立了以生理学为基础的神经科学的雏形，对建立简单的数学模型来模拟神经元的动态行为已经具有一定的启发意义。

生理学

Physiology

- ▶ At the very early stage of physiology, its boundary between philosophy, psychology and medicine was not that clear. And it's more like an assistive method for other disciplines. For example, by better understanding of human body's to construct a mechanical apparatus to carry out cognitive operations to verify materialism. To understand how brains store and process information in order to advance the study of psychology. To dissect and study the human body to discover better approaches to cure people.
- ▶ However, the development of technologies and enrichment of experimental methods promoted the study of physiology. For example, in neural-physiology, new ways of dyeing and microscopic in central nervous system and peripheral nervous system study led to the establishment of brain's fundamental role for cognition, and conclusion of collective behaviours of simple neurons inducing reasoning and actions.
- ▶ In addition, some scholars, such as Hermann von Helmholtz and Wilhelm Wundt, employed scientific methods to investigate the human visual system.
- ▶ By the 1940s, the foundations of neuroscience based on physiology had largely taken shape, providing inspiration for the subsequent development of simple mathematical models to simulate the dynamic behavior of neurons.

控制论

Control Theory

- 工具的适用大大延拓了人类的活动范围，促进了人类文明的发展，因此控制很早就进入了人们的生活，如亚历山大时代，就发明了带有调节器的水钟，称为克泰西比奥斯（Ctesibius）。
- 随着对控制的理论研究与实践应用，人们也较早地认识到，控制是服务于有目的行为的一种调节机制，旨在最小化目标状态与当前状态之间的差异（“误差”）。
- 第一次工业革命与第二次工业革命的发展，控制理论逐渐成为工程领域的核心概念，其用于设计和优化各种系统，包括机械、电子、通信、航空等系统，以实现预期的性能和效果。
- 随着时代发展与场景复杂化，工程师不仅期望设计出的系统在给定的条件下具有稳定的控制行为，还期望能自动适应变化的条件，实现自动调节，即具有一定的智能性。这也促使工程界加入对智能的研究。

控制论

Control Theory

- ▶ The usage of tools has greatly expanded people's exploring range and promoted civilization. Therefore, control entered into people's daily life at a very early stage. For example, in the era of Alexander, people invented the water clock with regulators called Ctesibius.
- ▶ With the research and applications of cybernetics, people realized that control is an adjustment for a specific purpose, by minimizing the difference between expected state and current state.
- ▶ The First Industrial Revolution and Second Industrial Revolution make control theory the core idea of engineering fields, to design and optimize all sorts of systems, including mechanics, electronics, communication, navigation systems, etc., to realize the expected performance.
- ▶ The sophistication of scenarios push the engineers to design systems not only with robust behavior under control, but also endowed with the ability to adapt to the changing conditions. This also pushes industries to join the trends for intelligence research.

其它方面

Miscellaneous

- ▶ 人们对通用可编程计算设备的研究已经有较长时间的历史：
 - ▶ 如在17世纪中期，由帕斯卡发明的帕斯卡计算器，可进行机械的加法和减法，后来，莱布尼兹又进行改良，增加了乘法。
 - ▶ 查尔斯·巴贝奇（1792-1871）尝试发明一种分析机进行通用计算，虽然未完成，但提出的可寻址内存、存储程序、条件跳转等思想，艾达·洛夫莱斯基于分析机的工作，都为后人的工作提供了宝贵的经验。
 - ▶ 上世纪初，关于这方面的工作有了显著进展，艾伦·图灵于1940年在英国发明了用于破译密码的数字电子计算机；康拉德·祖斯于1941年在德国研制了Z-3，第一台可编程计算机的工作；约翰·阿塔纳索夫于1940-42年在美国研制了第一台电子计算机ABC；约翰·莫奇利和约翰·埃克特于1945年在美国领导了ENIAC的工作，实现了第一台通用、电子、数字计算机。
- ▶ 总之，计算机技术迅速发展，计算机硬件和软件的进步，使得人们开始思考如何利用计算机来模拟人类的智能。

其它方面

Miscellaneous

- ▶ The exploration of general-purpose programmable computing devices has a long history.
 - ▶ For example, in the mid-17th century, Pascal's calculator (the Pascaline), invented by Blaise Pascal, was capable of performing mechanical addition and subtraction. Later, Gottfried Wilhelm Leibniz improved upon this design, enabling multiplication and division operations.
 - ▶ In the 19th century, Charles Babbage attempted to design the Analytical Engine for general-purpose computation. Although the machine was never completed, its concepts—such as addressable memory, stored programs, and conditional control—as well as the work of Ada Lovelace based on the Analytical Engine, provided valuable foundations for later developments in computing.
 - ▶ By the first half of the 20th century, substantial progress had been made in the development of computing devices. Alan Turing participated in research related to cryptographic computing systems in the United Kingdom during the 1940s. Konrad Zuse developed the Z3 in Germany in 1941, which is widely regarded as one of the earliest programmable automatic digital computers. John Vincent Atanasoff contributed to the development of the ABC (Atanasoff–Berry Computer) between 1940 and 1942 in the United States, an important early electronic digital computer. In 1945, John Mauchly and J. Presper Eckert led the development of ENIAC, marking a major milestone in general-purpose electronic digital computing.
- ▶ In summary, the rapid advancement of computer technology, together with continual progress in computer hardware and software, prompted people to consider how computers might be used to simulate human intelligence.

其它方面

Miscellaneous

- ▶ 一些学科新出现或根据当时的学术成就被重新阐释：
 - ▶ 克劳德·香农创立了信息论，虽然信息论旨在为数据通信的基本限制以及信息的高效利用提供一个数学框架，但一些概念，例如熵等，为本学科及其它学科的发展提供了深刻的见地。
 - ▶ 约翰·冯·诺依曼和奥斯卡·摩根斯特恩结合概率论与效用理论创立了决策理论，这和行为心理学等学科结合，为后来人工智能的发展提供了积极意义。
 - ▶ 艾伦·图灵人工智能和机器智能的深刻思考，撰写了注明的《计算机与智能》一文，首次提出图灵测试，这大大引发了人们创建一个系统通过图灵测试的兴趣。
- ▶ 总之，围绕计算与智能的议题达到一种临界状态，促使人们以更明晰的方式定义一种新的学科及其研究对象。

其它方面

Miscellaneous

- ▶ Some emerging disciplines began to take shape, while existing fields were reinterpreted in light of contemporary academic developments.
 - ▶ Claude Shannon established the foundations of information theory. Although information theory was primarily intended to provide a mathematical framework for the fundamental limits of data communication and the efficient transmission and utilization of information, several of its core concepts, such as entropy, provided profound insights for the development of many disciplines, including artificial intelligence.
 - ▶ John von Neumann and Oskar Morgenstern combined probabilistic thinking with utility theory, laying important foundations for modern game theory and decision analysis. Together with developments in fields such as behavioral psychology and economics, these ideas positively influenced the subsequent development of artificial intelligence.
 - ▶ Alan Turing engaged in profound reflections on artificial and machine intelligence and authored the influential paper *Computing Machinery and Intelligence*. In this work, he systematically proposed the idea that later became known as the Turing Test, greatly stimulating interest in constructing systems capable of exhibiting human-like intelligence.
- ▶ Overall, research surrounding computation and intelligence gradually reached a critical stage, prompting scholars to define a new discipline and its objects of study in a clearer and more systematic manner.

新生伊始（1943~1956）

Origination (1943~1956)

- ▶ 沃伦·麦卡洛和沃尔特·皮茨参照当时对神经元的研究结论，首次引入了人工神经元的概念，他们使用的是开关型或二值值的神经元，并展示了这样一组神经元组成的人工神经网络可以用于表示逻辑函数。这些先驱性工作为人工神经网络奠定了基础。
- ▶ 开关型的神经元一个局限是连接神经元的权重是固定的，为了模拟神经元的可塑性，人们开始了学习方法的研究。是唐纳德·赫布在1949年提出了一个学习规则，称赫布学习规则或赫布法则。
- ▶ 人们逐步考虑在一些形式化问题，如棋类游戏，数学逻辑中引入人工智能，例如，图灵与香农分别在1950年左右为国际象棋设计了程序，以展示机器模拟人类思维和问题解决的潜力。
- ▶ 马文·明斯基和迪恩·埃德蒙兹于1951年构建了第一个人工神经网络计算机SNARC（随机神经模拟增强计算器），它旨在使用电子元件模拟简单的神经行为。

新生伊始 (1943~1956)

Origination (1943~1956)

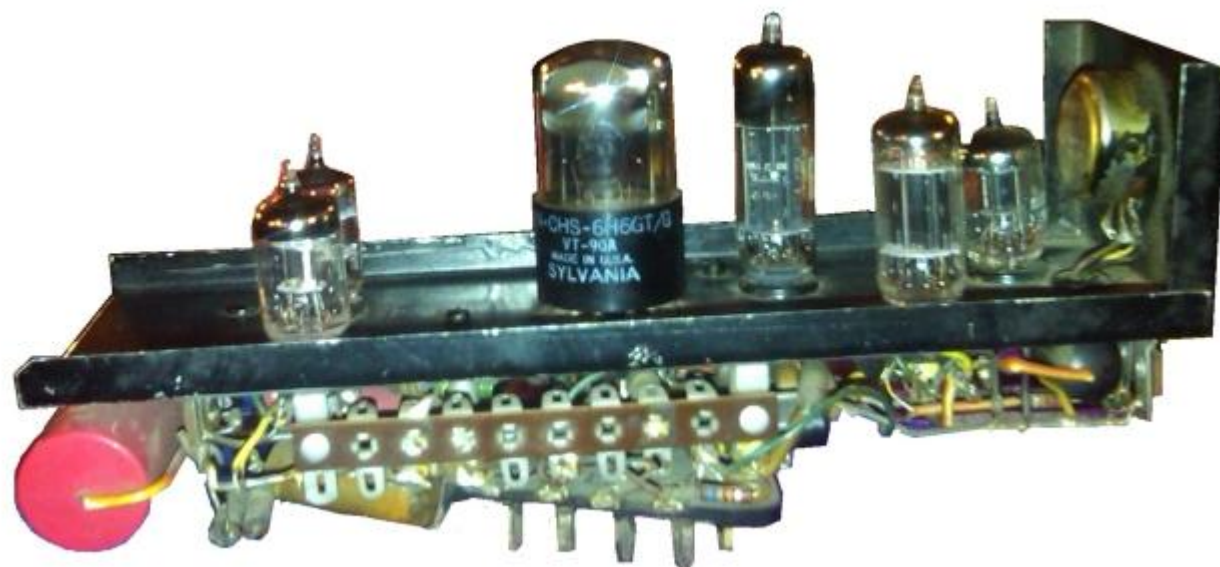
- ▶ Warren McCulloch and Walter Pitts drew upon contemporary findings in neuroscience and introduced one of the earliest artificial neuron models. They employed switch-based (binary) neurons and demonstrated that artificial neural networks composed of such neurons could represent logical functions. These pioneering efforts laid the foundation for the development of artificial neural networks.
- ▶ However, one limitation of switch-based neurons was the fixed connection weights between neurons. To simulate neural plasticity, researchers began to investigate learning mechanisms. In 1949, Donald Hebb proposed a learning rule that later became known as the Hebbian learning rule, or Hebb's rule.
- ▶ Researchers also gradually explored the application of artificial intelligence to formalized problems, such as board games and mathematical logic. For example, around the 1950s, Alan Turing and Claude Shannon independently designed chess-playing programs to demonstrate the potential of machines to simulate human reasoning and problem-solving.
- ▶ In 1951, Marvin Minsky and Dean Edmonds constructed SNARC (Stochastic Neural Analog Reinforcement Calculator), an early artificial neural network computer designed to simulate simple neural behavior using electronic components.

新生伊始 (1943~1956)

Origination (1943~1956)

- ▶ 马文·明斯基 (1927-2016) 是美国的一位计算机科学家和认知科学家，他是人工智能领域的先驱之一。他于1927年8月9日出生在美国纽约市，很早就显示了对数学和科学的兴趣。他在哈佛大学获得数学学士学位，随后在普林斯顿大学获得数学博士学位。
- ▶ 在学生时期，明斯基就梦想着制造出一种能够通过提供与“突触”相连的“神经元”内存来学习的机器，同时具备存储过去的记忆的功能，以便在面对不同情况时能够高效地运行。通过与迪恩·埃德蒙兹合作，他们在1951年构建了第一个人工神经网络计算机。
- ▶ 该神经计算机有40个神经元组成，名为SNARC (Stochastic Neural Analog Reinforcement Computer)。该神经计算机采用了赫布学习法，突触根据执行特定任务的成功程度来调整其权重（突触渗透性的度量）。该机器由管子、马达和离合器构建，成功地模拟了在迷宫中搜索食物的老鼠行为。

模拟神经元 →



新生伊始 (1943~1956)

Origination (1943~1956)

- ▶ Marvin Minsky (1927–2016) was an American computer scientist and cognitive scientist, and one of the pioneers of artificial intelligence. Born in New York City on August 9, 1927, he demonstrated a strong interest in mathematics and science from an early age. He received a bachelor's degree in mathematics from Harvard University and later earned a Ph.D. in mathematics from Princeton University.
- ▶ During his student years, Minsky envisioned building a machine capable of learning. Such a system would store experience through a network analogous to neurons and synapses and retain memories of past events, enabling it to respond more effectively to different situations. In 1951, together with Dean Edmonds, he constructed SNARC, an early artificial neural network computer.
- ▶ SNARC (Stochastic Neural Analog Reinforcement Calculator) consisted of approximately forty artificial neurons and employed ideas inspired by Hebbian learning, adjusting connection weights based on feedback from task performance. Built using vacuum tubes, motors, and clutches, the machine successfully simulated the behavior of a rat searching for food in a maze.

新生伊始（1943~1956）

Origination (1943~1956)

- ▶ 1956年夏天，由约翰·麦卡锡、马文·明斯基、纳撒尼尔·罗切斯特和克劳德·香农在美国新罕布什尔州汉诺威市的达特茅斯学院组织了一个研讨会（达特茅斯夏季人工智能研究项目），其主要目标是汇集来自不同领域的研究人员，探讨创造能够展示智能行为的机器的可能性。达特茅斯研讨会被认为是人工智能历史上的一个里程碑事件，取得了一系列的成果：
 - ▶ “人工智能”术语的提出：在研讨会期间首次提出了“人工智能”这个术语，标志着这个术语首次被用来描述这一领域。（这个术语一般认为是经由约翰·麦卡锡提出，会议讨论确定的）
 - ▶ 人工智能学科的形成：该研讨会将人工智能作为一个独特的科学和学术学科进行了正式界定，强调了创造智能机器和系统的研究。
 - ▶ 确定了人工智能研究领域：参与者确定了人工智能领域内的几个关键研究领域，包括问题解决、符号推理、自然语言处理、神经网络和机器学习。
 - ▶ 早期人工智能程序和概念：参与者讨论和展示了早期人工智能程序和概念，其中包括符号操作、基于规则的系统，以及模仿人类认知过程的方法。

新生伊始 (1943~1956)

Origination (1943~1956)

- ▶ In the summer of 1956, John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon organized the Dartmouth Summer Research Project on Artificial Intelligence at Dartmouth College in Hanover, New Hampshire, United States. The primary objective of the meeting was to bring together researchers from different disciplines to explore the possibility of constructing machines capable of exhibiting intelligent behavior. The Dartmouth conference is widely regarded as a landmark event in the history of artificial intelligence, and its influence can be summarized in several major aspects:
 - ▶ Introduction of the term “Artificial Intelligence”: The term *Artificial Intelligence* was formally introduced in the conference proposal, marking the beginning of its use to describe this emerging field of study. The term is generally credited to John McCarthy and was accepted during the planning and discussions surrounding the conference.
 - ▶ Formation of artificial intelligence as a field: The Dartmouth conference promoted the development of artificial intelligence as a distinct area of research, gradually shaping it into an independent academic discipline focused on the construction of intelligent machines and systems.
 - ▶ Establishment of major research themes in AI: Participants discussed several key problems, including problem solving, symbolic reasoning, natural language processing, learning mechanisms, and neural networks—topics that later became important areas of artificial intelligence research.
 - ▶ Early AI programs and concepts: Participants exchanged ideas and discussed early AI programs and concepts, including symbolic manipulation, rule-based reasoning systems, and approaches aimed at simulating human cognitive processes.

新生伊始（1943~1956）

Origination (1943~1956)

- ▶ 对人工智能的热情：研讨会在与会者中引发了对人工智能可能性的热情和兴奋，以及它对语言翻译、逻辑推理和模式识别等领域的潜在影响。
- ▶ 跨学科合作：该研讨会促进了跨学科合作，将数学、计算机科学、认知科学、心理学等领域的研究人员汇集在一起。这种合作有助于塑造多方位的人工智能研究。
- ▶ 人工智能作为人类智能的模拟：研讨会的讨论围绕着创建能够模拟人类智能和决策的系统，启发了人工智能研究中旨在复制人类思维过程的研究方向。
- ▶ 总之，1956年的达特茅斯研讨会标志着人工智能领域的正式开始。它为随后的研究、讨论和进步奠定了基础，塑造了几十年来人工智能技术和应用的发展。
- ▶ 同时期的工作还有艾伦·纽厄尔、赫伯特·A·西蒙和克利夫·肖于1956年关于逻辑推理者（Logic Theorist）的工作。它是第一个有意识地设计用于自动推理的计算机程序，被描述为“第一个人工智能程序”。它证明了怀特海德和罗素的《数学原理》第二章中前52个定理中的38个，还为其中一些定理找到了新的、更短的证明。

新生伊始 (1943~1956)

Origination (1943~1956)

- ▶ Enthusiasm for artificial intelligence research: The Dartmouth conference generated enthusiasm and optimism among participants regarding the potential of artificial intelligence and its possible applications in areas such as language translation, logical reasoning, and pattern recognition.
- ▶ Interdisciplinary collaboration: The conference promoted interdisciplinary collaboration by bringing together researchers from mathematics, computer science, psychology, cognitive science, and related fields. This collaborative environment contributed to the diversification of AI research and gradually fostered a research paradigm integrating multiple disciplinary perspectives.
- ▶ Artificial intelligence as a simulation of human intelligence: The conference also proposed the construction of systems capable of simulating human intelligence and decision-making. This perspective inspired subsequent lines of AI research aimed at modeling or reproducing aspects of human cognitive processes.

Overall, the Dartmouth conference of 1956 is widely regarded as an important starting point in the history of artificial intelligence. It laid the foundation for subsequent research, discussions, and technological progress, profoundly shaping the development of AI theories, technologies, and applications over the following decades.

新生伊始 (1943~1956)

Origination (1943~1956)

- ▶ The contemporary development includes the Logic Theorist program created in 1956 by Allen Newell, Herbert A. Simon, and Cliff Shaw. The program is generally regarded as the first computer program explicitly designed for automated reasoning and is often described as “the first artificial intelligence program.” It successfully proved 38 of the first 52 theorems in Chapter 2 of *Principia Mathematica*, written by Alfred North Whitehead and Bertrand Russell, and even discovered shorter proofs for some of them.

- ▶ 下面的Python代码展示了一段简单的逻辑推理程序：

The following code snippet demonstrates a simple logic-based reasoning program:

```
from sympy.logic.boolalg import And, Or, Not
from sympy.abc import A, B
expr = And(A, Or(B, Not(A)))
result = expr.simplify()
print(f"Simplified logic expression: {result}")
```

徐徐发展（1956~1969）

Development (1956~1969)

- ▶ 弗兰克·罗森布拉特于1957年开发了感知器或感知机，是早期的人工神经网络的典型代表。感知器是一种简单的人工神经元网络，其灵感来自于生物神经元的工作原理。它由输入层和一个带有权重的输出神经元组成。每个输入与权重相乘后的结果被送入输出神经元，并经过一个阈值函数，最终输出结果。罗森布拉特在他的工作中引入了一些关键概念，其中一些为当今神经网络的基础所在：
 - ▶ 权重和阈值：感知器的核心概念是权重和阈值。权重表示输入特征的重要性，而阈值决定了神经元是否激活。
 - ▶ 学习算法：罗森布拉特提出了一种称为“感知器规则”的学习算法，通过调整权重来使感知器逐步适应训练数据。这被认为是梯度下降算法的一个早期形式。
- ▶ 约翰·麦卡锡于1958年开发了编程语言Lisp，它成为了在人工智能研究中最广泛使用的编程语言。

徐徐发展 (1956~1969)

Development (1956~1969)

- ▶ Frank Rosenblatt introduced and developed the Perceptron model in 1957, an early representative of artificial neural networks. The perceptron is a simple artificial neural network inspired by the information-processing mechanisms of biological neurons. It consists of an input layer and an output neuron with adjustable weights. Input signals are multiplied by their corresponding weights, then passed through an activation function or a threshold function after summation to generate the final output. In his work, Rosenblatt introduced several key ideas, several of which later underpin the modern neural networks:
 - ▶ Weights and thresholds: they are central concepts of the perceptron. Weights represent the relative importance of input features, while thresholds determine whether a neuron is activated.
 - ▶ Learning algorithm: Rosenblatt proposed a learning method called Perceptron Learning Rule, in which weights are iteratively adjusted to enable the model to adapt to training data. This idea is widely regarded as an important early exploration of later gradient-based optimization approaches.
- ▶ In addition, John McCarthy developed the programming language Lisp in 1958. Lisp later became one of the most widely used programming languages in artificial intelligence research and played an important role in the development of symbolic AI.

徐徐发展（1956~1969）

Development (1956~1969)

- ▶ 阿瑟·塞缪尔于1959年创造了“机器学习”这一术语。塞缪尔主要工作是通过游戏，特别是跳棋游戏，致力于开发可以自我学习的计算机程序。机器学习的特点和途径是使计算机通过从数据中学习来改进自己的性能。这个术语的提出标志着机器学习作为一门重要的学科的形成。
- ▶ 伯纳德·威德罗和泰德·霍夫在1960年设计了ADALINE，随后研究者在ADALINE的基础上开发了MADALINE，其为三层（输入层、隐藏层、输出层）的全连接的前馈人工神经网络。
- ▶ 弗兰克·罗森布拉特在1962年提出并证明了感知机收敛定理，即感知机在线性可分情况下能够学习出一个能够完美分类训练样本的决策边界。
- ▶ 1969年，马文·明斯基和西摩·帕普特发表了《Perceptrons: An Introduction to Computational Geometry》，强调了简单神经网络的局限性，对神经网络在当期的发展有一定的影响。

徐徐发展 (1956~1969)

Development (1956~1969)

- ▶ Arthur Samuel introduced the term “Machine Learning” in 1959. Samuel’s research focused primarily on developing computer programs capable of self-improvement through games, particularly checkers. The central idea of machine learning is to enable computers to learn from data or experience and progressively improve their performance. The introduction of this term marked the gradual emergence of machine learning as an independent research direction within artificial intelligence.
- ▶ Bernard Widrow and Marcian Hoff (Ted Hoff) designed ADALINE (Adaptive Linear Neuron) in 1960. Subsequently, researchers developed MADALINE (Multiple ADaptive LINear Elements) based on ADALINE, an early form of feedforward multilayer artificial neural networks.
- ▶ Frank Rosenblatt proposed and proved the Perceptron Convergence Theorem in 1962, demonstrating that a decision boundary is reachable during training when training samples are linearly separable.
- ▶ In 1969, Marvin Minsky and Seymour Papert published the book *Perceptrons: An Introduction to Computational Geometry*, which elaborated the limitations of simple perceptron models. This work had some influence on artificial neural network research.

徐徐发展（1956~1969）

Development (1956~1969)

- ▶ 休伯特·德雷福斯曾于1965年发表了名为“Alchemy and AI”的论文，主张人类思维并非如同计算机，同时认为人工智能在某些界限之下将无法继续进展。这为神经网络后续的发展埋下了悲观的基调的伏笔。
- ▶ 马文·明斯基和西摩·帕普特所著的《Perceptrons: An Introduction to Computational Geometry》一书中，有著名的“异或门”问题，即单层感知机不能解决XOR问题。尽管作者仅暗示单个神经元无法实现某些函数，如异或逻辑函数，但一些不恰当的解读被引申，即较大的网络也可能存在类似的局限性，因此应该被放弃。
- ▶ Hubert Dreyfus published the paper *Alchemy and Artificial Intelligence* in 1965, argued that human thinking could not be simply regarded as a process analogous to symbolic computation performed by computers and suggested that the progress of artificial intelligence might face fundamental limitations. To some extent, these views reinforced skepticism toward AI research which attributed to the decline in neural network research.
- ▶ In the book “Perceptrons: An Introduction to Computational Geometry”, Marvin Minsky and Seymour Papert discussed the well-known XOR (exclusive OR) problem, demonstrating that a single-layer perceptron cannot represent linearly inseparable functions such as the XOR logical function. Although the authors mainly emphasized the limitations of single-layer perceptrons, some researchers interpreted the work pessimistically, concluding that larger neural networks might suffer from similar limitations, indicating refraining from neural networks.

徐徐发展（1956~1969）

Development (1956~1969)

- ▶ 在人工智能的发展演进过程中，涌现了多个不同的学派，如连接主义、符号主义等。
 - ▶ 连接主义学派强调模仿生物神经网络的方式来构建智能系统。它将大量简单的神经元相互连接，通过学习调整连接权重来实现智能行为。这种学派关注于基于数据的学习，如神经网络等技术，强调从经验中获取知识和能力。
 - ▶ 符号主义学派认为智能行为可以通过符号和规则的处理来实现。它将知识表示为符号，通过推理和规则运算来实现问题解决。这种学派强调逻辑推理、专家系统等方法，注重形式化的知识表示和推理。
- ▶ During the development of artificial intelligence, several influential schools gradually emerged, among which connectionism and symbolicism became particularly prominent.
 - ▶ Connectionism: Connectionism emphasizes the construction of intelligent systems by modeling biological neural systems. This approach typically employs networks composed of a large number of interconnected simple processing units, where intelligence is achieved by adjusting connection weights through learning. Connectionism represented by artificial neural networks highlights learning from data and experience.
 - ▶ Symbolicism: Symbolicism argues that intelligent behavior can be achieved through the representation and manipulation of symbols and rules. It typically represented knowledge in symbolic form, to solve problems by logical reasoning and rule matchin. Symbolicism is represented by approaches such as logical inference, knowledge representation, and expert systems, emphasizing explicit knowledge representation, formal reasoning, and interpretability.

徐徐发展（1956~1969）

Development (1956~1969)

- ▶ 实际上，在1960年代，符号主义学派在人工智能的发展中占据主导地位。这个学派强调使用符号和逻辑推理来表示和处理知识，它在推理、问题求解和专家系统等领域取得了显著进展。这个时期的人工智能研究主要集中在符号主义的范畴内。这在自然语言处理的研究上反映更为明显：
 - ▶ 丹尼尔·鲍布罗于1964年完成了他在麻省理工学院的博士学位论文，题为《用于计算机问题解决系统的自然语言输入》，并开发了STUDENT，一个自然语言理解计算机程序。
 - ▶ 约瑟夫·韦茨恩鲍姆于1965年开发了ELIZA，一个交互式程序，可以用英语就任何主题进行对话。
 - ▶ 特里·温诺格拉德于1968年开发了SHRDLU，一款早期的自然语言理解计算机程序。
- ▶ In fact, during the 1960s, artificial intelligence research is dominated by symbolism. This school emphasized the representation of knowledge through symbols and the realization of intelligent behavior through logical reasoning, rule-based operations, and search mechanisms, and achieved significant progress in areas such as reasoning, problem solving, and expert systems. Consequently, AI research during this period was largely guided by symbolic approaches. This trend was particularly evident in natural language processing (NLP) research:
 - ▶ Daniel Bobrow completed his doctoral dissertation at the Massachusetts Institute of Technology in 1964, titled *Natural Language Input for a Computer Problem Solving System*, and developed STUDENT, an early natural language processing program capable of understanding and solving mathematical word problems.
 - ▶ Joseph Weizenbaum developed ELIZA in 1965, an interactive natural language dialogue system capable of simple English conversations across various topics.
 - ▶ Around 1968, Terry Winograd developed SHRDLU, one of the earliest representative natural language understanding systems.

波澜不惊（1969~1979）

Stagnation (1969~1979)

- ▶ 由于早期符号主义的盛行，早期的人工智能成果的另一个典型代表还包括爱德华·费根鲍姆、布鲁斯·G·布坎南、乔舒亚·莱德伯格和卡尔·杰拉西于1965年在斯坦福大学开发的DENDRAL项目。这是第一个专家系统，它自动化了有机化学家的决策过程和问题解决行为，旨在研究假设形成和构建科学中经验归纳的模型。
- ▶ 基于规则形式化方法会不可避免地导致计算可行性的研究。由于人工智能处理的问题在数学上皆是非线性问题，当问题的复杂性，即难以用有效的算法在合理的时间内解决时，“可解性”或“不可解性”必将成为计算机科学和人工智能领域的一个重要的研究方向。这一时期已知一些问题可能没有已知的高效解决方法，解决过程通常涉及大量的计算和搜索，解决方案的时间和资源成本非常高；一些问题可能有解决方法，即使在算法设计、资源分配等进行多方面的优化工作，但需要非常长的时间来完成。
- ▶ “不可解性”导致了人工智能的中期发展特别是从形式化方法、神经网络的角度来讲的一个低潮期，在人们寻找多样化方法的同时，仿生计算逐步发展起来。

波澜不惊 (1969~1979)

Stagnation (1969~1979)

- ▶ Another important achievement of early symbolic artificial intelligence was the DENDRAL project, developed in 1965 at Stanford University by Edward Feigenbaum, Bruce Buchanan, Joshua Lederberg, and Carl Djerassi. DENDRAL is generally regarded as one of the earliest successful expert systems. Its goal was to assist organic chemists in molecular structure inference, automating aspects of scientific reasoning and problem solving while also exploring mechanisms of hypothesis formation and inductive reasoning in scientific discovery.
- ▶ Formal rule-based approaches inevitably led to studies of computational feasibility. Actually, impractical potential solutions of nonlinear AI problems cause issues such as computability, tractability, and computational complexity became important topics in artificial intelligence. Researchers recognized that some theoretically-solvable problems required extensive computation thus incurring substantial time and resource costs. Other solvable problems through optimized algorithms, heuristic search, or improved resource allocation, could still require prohibitively long execution times.
- ▶ Awareness of computational limitations and the constraints of formal approaches influenced the direction of AI research and contributed, to some extent, to the period of reduced momentum in AI development during the 1970s and early 1980s, often referred to as the first AI winter. At the same time, researchers began exploring more diverse methodologies, and bio-inspired computing as well as artificial neural network research gradually regained attention.

波澜不惊（1969~1979）

Stagnation (1969~1979)

- ▶ 斯坦福大学于1972年开发了MYCIN，一个早期的专家系统。该系统用于识别引起严重感染的细菌并推荐抗生素。这个系统的知识根据传染病专家的临床决策标准，以约350条规则的形式进行编码，辅助系统的推理过程。
- ▶ 卡内基梅隆大学于1978年开发了XCON（eXpert CONfigurer）程序，这是一个基于规则的专家系统，用于通过根据客户需求自动选择组件，辅助于订购DEC的VAX计算机。
- ▶ 约翰·荷兰在1960年代提供了进化计算的概念并于60年代末和70年代初，首次引入了遗传算法的概念，并在此基础上发展了一系列与进化计算相关的技术和方法。他的著作“Adaptation in Natural and Artificial Systems”于1975年出版，详细阐述了遗传算法和进化计算的理论基础和应用。
- ▶ 对基于规则的专家系统或人工智能系统，知识表示是使计算机能够获取、存储、操作和推理知识的关键环节。通俗地讲，知识表示指的是将人类知识以计算机可以理解和处理的方式进行编码和存储的过程，这也是这一时间的一个研究热点，尽管可能没有达到预期的效果。

波澜不惊 (1969~1979)

Stagnation (1969~1979)

- ▶ Stanford University developed MYCIN in 1972, an influential early expert system to identify bacteria responsible for severe infections and recommend antibiotic treatments. Its knowledge base was constructed using approximately 350 encoded rules derived from the clinical experience and decision-making standards of infectious disease specialists, thereby supporting the system's reasoning and diagnostic processes.
- ▶ In 1978, Carnegie Mellon University developed XCON (eXpert CONfigurer), a rule-based expert system designed to assist in the configuration and ordering of VAX computers produced by Digital Equipment Corporation (DEC).
- ▶ John Holland proposed evolutionary computation during the 1960s and systematically introduced the concept of the genetic algorithm in the late 1960s and early 1970s. His 1975 book, *Adaptation in Natural and Artificial Systems*, provided a detailed account of the theoretical foundations and applications of genetic algorithms and evolutionary computation.
- ▶ For rule-based expert systems and early artificial intelligence systems, knowledge representation constituted a crucial component enabling computers to acquire, store, organize, manipulate, and reason about knowledge. In simple terms, knowledge representation refers to the process of formally encoding and storing human knowledge in a form that computers can understand and process, a major focus of AI research during that period. Although notable progress was achieved, the results still fell short of the optimistic expectations held by many researchers.

波澜不惊（1969~1979）

Stagnation (1969~1979)

- ▶ 知识表示涉及将现实世界中的实体、事实、概念和关系转化为计算机可以处理的数据结构，如逻辑表达式、语义网络、框架、本体等。不同的知识表示方法适用于不同类型的问题和应用领域。
- ▶ 常见的知识表示方法包括：
 - ▶ 逻辑表示：使用逻辑语言表示知识，如一阶逻辑、描述逻辑等，可以用于推理和推断。
 - ▶ 语义网络：通过节点和连接表示概念和关系，适用于表示语义关系。
 - ▶ 框架表示：使用框架或类似的结构表示对象的属性、关系和行为。
 - ▶ 本体表示：使用本体语言表示概念、实体和关系，用于构建共享的知识库。
- ▶ 知识表示在人工智能的各个领域中都起着关键作用，如专家系统、自然语言处理、机器学习等。有效的知识表示方法可以帮助计算机更好地理解 and 处理人类知识，从而实现更智能的决策和推理。

波澜不惊 (1969~1979)

Stagnation (1969~1979)

- ▶ Knowledge representation involves transforming entities, facts, concepts, and relationships in the real world into formal structures that computers can process, such as logical expressions, semantic networks, frames, and ontologies. Different knowledge representation methods are suitable for different types of problems and application scenarios, and therefore appropriate representations are typically selected according to specific tasks in artificial intelligence systems.
- ▶ Common approaches to knowledge representation include:
 - ▶ Logical Representation: Knowledge is represented using formal logical languages, such as first-order logic and description logic. These approaches are suitable for tasks such as reasoning and inference.
 - ▶ Semantic Networks: Concepts and semantic relationships are represented through nodes and connections. This approach is particularly suitable for representing hierarchical structures and semantic associations.
 - ▶ Frame Representation: Objects are represented using frame-like data structures that describe attributes, relationships, and behavioral characteristics, making them effective for representing hierarchical knowledge and default properties.
 - ▶ Ontology Representation: Ontology languages are used to formally describe concepts, entities, and their relationships in order to construct knowledge systems with shared semantics.
- ▶ Knowledge representation plays a crucial role across many areas of artificial intelligence, including expert systems, natural language processing, knowledge engineering, and machine learning. Effective knowledge representation methods help computers more accurately understand, organize, and utilize knowledge, thereby improving decision-making, reasoning, and problem-solving capabilities.

重装上阵（1980~1990）

Reloaded (1980~1990)

- ▶ 实际上，神经网络的一个应用的局限是学习方法的问题。尽管对于简单的神经网络，基于赫布法则的感知机学习方法可以调整简单网络的权重，但对于深层网络却力不从心。后面又出现了局部扰动法则，由于没有全局的指导，导致该方法效率低下。
- ▶ 在70年代，多位研究者便开始探索通过误差传播来训练神经网络，如保罗·韦尔博斯和大卫·帕克独立地发展了与反向传播类似的思想，特别是保罗·韦尔博在1974年的博士论文中发表了与此相关的工作。
- ▶ 大卫·鲁梅尔哈特、杰弗里·辛顿和罗纳德·威廉姆斯于1986年10月发表了“Learning representations by back-propagating errors”一文，文中他们描述了“一种新的学习过程，即反向传播，适用于神经元样单元的网络。杰弗里·辛顿的工作对反向传播的推广和对于其在神经网络领域更广泛认可和应用做出了重要贡献。
- ▶ 乔治·赛本科于1989年证明了通用逼近定理，即在一定条件下，一个足够大的神经网络可以以任意精度逼近任意连续函数。其为神经网络在函数逼近和模式识别等任务上的广泛应用提供了理论基础。这个定理的存在也促使人们对神经网络的深度和宽度进行了更深入的研究，为深度学习的兴起铺平了道路。

重装上阵 (1980~1990)

Reloaded (1980~1990)

- ▶ In fact, one important limitation in the early development of artificial neural networks lay in the inadequacy of available learning algorithms. For simple neural networks, perceptron learning methods inspired by Hebbian learning principles could adjust connection weights and accomplish relatively simple classification tasks. However, as network depth increased, these methods became ineffective for training deep architectures. Researchers subsequently explored several local learning strategies, such as local perturbation methods, but because these approaches lacked effective guidance from global error information, their efficiency remained limited and they proved difficult to scale to complex networks.
- ▶ During the 1970s, several researchers began to investigate methods for training neural networks through error propagation. For example, Paul Werbos and David Parker independently developed ideas similar to backpropagation. In particular, Werbos systematically discussed the use of the chain rule to propagate errors for training multilayer networks in his 1974 doctoral dissertation, a contribution now widely regarded as an important theoretical foundation of the backpropagation algorithm.

重装上阵 (1980~1990)

Reloaded (1980~1990)

- ▶ In 1986, David Rumelhart, Geoffrey Hinton, and Ronald J. Williams published the landmark paper Learning Representations by Back-Propagating Errors, which systematically described the backpropagation algorithm for training multilayer neural networks. The method enables networks to propagate errors backward from the output layer toward earlier layers, thereby adjusting connection weights and achieving effective learning. In particular, Hinton played an important role in promoting the broader recognition and application of backpropagation in neural network research.
- ▶ George Cybenko proved the Universal Approximation Theorem in 1989, showing that under certain conditions, a feedforward neural network with a single hidden layer and sufficiently many neurons can approximate any continuous function defined on a compact domain with arbitrary precision. This theorem provided an important theoretical foundation for applying neural networks to tasks such as function approximation, pattern recognition, and regression, while also motivating deeper investigations into the influence of network width and depth on representational capacity, thereby paving the way for the later emergence of deep learning.

重装上阵（1980~1990）

Reloaded (1980~1990)

- ▶ 在此期间，人工智能在工业界也有所建树。如在1981年，日本国际贸易与工业部拨款8.5亿美元用于第五代计算机项目。该项目旨在开发能够进行对话、翻译语言、解释图片并像人类一样进行推理的计算机。在1986年，恩斯特·迪克曼斯指导下在慕尼黑的联邦国防军大学制造了首辆搭载摄像头和传感器、基于梅赛德斯-奔驰面包车的无人驾驶汽车。
- ▶ 1988年，裘德亚·珀尔发表了《Probabilistic Reasoning in Intelligent Systems》，奠定了处理不确定性信息的表示和计算的基础。特别是贝叶斯网络，不仅影响了人工智能领域，还成为许多其他工程和自然科学领域的重要工具。
- ▶ 同期，自然语言处理也由第一代基于规则的方法，向第二代基于统计的方法转变。基于统计的方法，主要是通过分析大量语言数据来构建模型和预测。这些方法包括词频、语法统计和概率模型，例如隐马尔可夫模型和最大熵模型。尽管在处理语义和上下文方面仍有局限性，统计方法已经能够处理更复杂的语言变化，基于此方法的一些结果也展示了一定的优越性。
- ▶ 尽管在1984年的AAAI（美国人工智能协会）的年度会议上，罗杰·尚克和马文·明斯基预警了即将到来的“人工智能冬天”，但反向传播算法等工作还是将基于神经网络的研究重又拉回连接学派。

重装上阵 (1980~1990)

Reloaded (1980~1990)

- ▶ During this period, artificial intelligence also achieved notable progress in industrial and engineering applications. For example, in 1981, the Japanese Ministry of International Trade and Industry (MITI) launched the Fifth Generation Computer Systems Project (FGCS) with funding of approximately \$850 million. The project aimed to develop knowledge-based computer systems capable of natural language dialogue, machine translation, image understanding, and human-like reasoning in order to advance intelligent computing. In 1986, under the guidance of Ernst Dickmanns, the Bundeswehr University Munich developed one of the earliest experimental autonomous vehicles. Built on a Mercedes-Benz van platform and equipped with cameras and sensors, the vehicle demonstrated autonomous navigation and driving capabilities under constrained conditions.
- ▶ In 1988, Judea Pearl published the book Probabilistic Reasoning in Intelligent Systems, which established an important theoretical foundation for the representation and reasoning of uncertain knowledge. In particular, Bayesian networks became a powerful framework for modeling probabilistic dependencies, influencing not only artificial intelligence but also fields such as statistics, biomedicine, and engineering science.

重装上阵 (1980~1990)

Reloaded (1980~1990)

- ▶ At the same period, natural language processing (NLP) gradually transitioned from first-generation rule-based methods to second-generation statistical approaches. Statistical methods primarily relied on analyzing large-scale language data to construct predictive models. Representative approaches included word-frequency statistics, language models, probabilistic grammars, and probabilistic models such as Hidden Markov Models (HMMs) and Maximum Entropy (MaxEnt) models. Although these approaches still faced limitations in deep semantic understanding and contextual modeling, they demonstrated clear advantages over purely rule-based systems in adaptability and practical performance.
- ▶ Although Roger Schank and Marvin Minsky warned of potential stagnation in AI research during the 1984 conference of the Association for the Advancement of Artificial Intelligence (AAAI)—later regarded as an early sign of the coming AI winter—the success of the backpropagation algorithm, together with gradual improvements in computing power, renewed interest in neural-network-based approaches and contributed to a return toward connectionist research paradigms.

韬光养晦（1990~2000）

Incubation (1990~2000)

- ▶ 1989年，AT&T贝尔实验室的扬·勒库恩和其他研究人员成功地将反向传播算法应用于多层神经网络，识别手写的邮政编码。1998年，在对反向传播进行优化之后，他们又发表了关于将神经网络应用于手写识别的论文。但是由于硬件的限制，训练这样的网络耗时太久，因此并没有引起较大的关注。但毫无疑问，这些工作基本构成了深度学习的基石。
- ▶ 1997年，塞普·霍赫莱特和尤尔根·施密特胡伯提出了长短时记忆（LSTM）模型，这是一种用于今天的手写识别和语音识别的循环神经网络类型。同样，由于硬件的限制，这个工作在当时也没有得到足够的重视。
- ▶ 同期，专家系统与自然语言处理进一步发展。1995年，理查德·沃利斯开发了聊天机器人A.L.I.C.E（Artificial Linguistic Internet Computer Entity）。
- ▶ 1997年，IBM开发的深蓝计算机成为第一个击败在位世界国际象棋冠军的计算机国际象棋程序。深蓝在国际象棋领域的成功被认为是计算机人工智能发展的一个里程碑，也标志着计算机在特定领域的能力不断提升。
- ▶ 支持向量机（SVM）由于有较好的理论基础（基于模式识别和统计学相关理论）与可解释性，在机器学习领域扮演着重要角色，为许多复杂问题提供了强大的解决方案，比许多传统的人工智能方法有更广泛的应用。

韬光养晦 (1990~2000)

Incubation (1990~2000)

- ▶ In 1989, Yann LeCun and his collaborators at Bell Labs successfully applied the backpropagation algorithm to multilayer neural networks for handwritten postal code recognition. Later, in 1998, LeCun and colleagues published the landmark paper Gradient-Based Learning Applied to Document Recognition, which systematically demonstrated the application of convolutional neural networks (CNNs) to tasks such as handwritten digit recognition, based on further optimizations of network architectures and training methods. Due to the limited computational hardware and high training costs at the time, these approaches did not initially attract widespread attention. Nevertheless, these pioneering efforts are widely regarded as forming an important foundation of modern deep learning.
- ▶ In 1997, Sepp Hochreiter and Jürgen Schmidhuber proposed the Long Short-Term Memory (LSTM) model, a specialized type of Recurrent Neural Network (RNN) designed to alleviate the problems of vanishing and exploding gradients encountered in training conventional recurrent neural networks on long sequences. LSTM networks were later widely applied to speech recognition, handwriting recognition, machine translation, and natural language processing. Similar to early deep neural networks, this work did not receive sufficient attention at the time due to limitations in computational resources and data availability.

韬光养晦 (1990~2000)

Incubation (1990~2000)

- ▶ Meanwhile, expert systems and natural language processing continued to advance. In 1995, Richard Wallace developed the chatbot A.L.I.C.E. (Artificial Linguistic Internet Computer Entity), a rule-based conversational system that became a representative example of early open-domain dialogue systems.
- ▶ In 1997, Deep Blue, developed by IBM, became the first computer chess system to defeat a reigning world chess champion in an official match. In its competition against Garry Kasparov, Deep Blue demonstrated the remarkable capabilities of computers in highly complex specialized tasks. This event is generally regarded as an important milestone in the history of artificial intelligence and highlighted the potential for computers to surpass human experts in specific domains.
- ▶ During the same period, the Support Vector Machine (SVM) gradually emerged as one of the most important machine learning methods in the late 1990s, owing to its strong theoretical foundation—particularly in statistical learning theory—as well as its strong generalization ability and interpretability. SVMs demonstrated strong performance in classification, regression, and pattern recognition tasks and were widely applied for many years in areas such as text classification, image recognition, and bioinformatics.

黎明在望（2000~2010）

Dawning (2000~2010)

- ▶ 2006年，Geoffrey Hinton发表了“Learning Multiple Layers of Representation”一文。这篇论文包含了利用非监督学习进行逐层训练，以便训练深度神经网络的方法。同时，这篇论文也对深度网络的知识表征提出了新的见解。
- ▶ 2007年，李飞飞教授和普林斯顿大学的同事开始构建ImageNet，这是一个大型的带注释图像数据库，旨在评计算机视觉物体识别方向的模型与算法研究。
- ▶ NVIDIA于2007年发布了首个初始版本的CUDA，允许开发人员在NVIDIA的图形处理器上进行高效的并行计算，极大地扩展了图形处理器应用于通用计算的领域。
- ▶ 2009年，拉贾特·莱纳、阿南德·马德哈万和安德鲁·吴发表了题为“Large-scale Deep Unsupervised Learning using Graphics Processors”的论文，论述了将现代图形处理器的超过多核CPU的计算能力，应用于机器学习方法的适用性。
- ▶ 2009年，谷歌开始秘密开发无人驾驶汽车。在后来的2012年，它成为第一个通过美国内华达州的自动驾驶测试的车辆。
- ▶ 2010年，ImageNet大规模视觉识别挑战（ILSVRC）正式启动，其演变成随后纪念每年举办的人工智能物体识别赛事，见证了人工智能特别是深度学习的许多标志性事件。

黎明在望 (2000~2010)

Dawning (2000~2010)

- ▶ In 2006, Geoffrey Hinton and his collaborators published the paper *A Fast Learning Algorithm for Deep Belief Nets*, introducing a training method for deep neural networks based on layer-wise unsupervised pretraining. This work effectively alleviated the difficulties associated with training deep networks and laid an important foundation for the revival of deep learning. At the same time, related studies provided new insights into the hierarchical knowledge representation capabilities of deep networks.
- ▶ In 2007, Fei-Fei Li and her colleagues at Princeton University began constructing ImageNet, a large-scale manually annotated image database designed to advance object recognition models and algorithms in computer vision while providing a unified benchmark for visual learning tasks.
- ▶ In 2007, NVIDIA released an early version of CUDA (Compute Unified Device Architecture), enabling developers to perform efficient parallel computation on NVIDIA graphics processing units (GPUs). This technology significantly expanded the role of GPUs from graphics rendering to general-purpose high-performance computing (GPGPU) and later provided essential computational support for the rapid advancement of deep learning.

黎明在望 (2000~2010)

Dawning (2000~2010)

- ▶ In 2009, Rajat Raina, Anand Madhavan, and Andrew Ng published the paper *Large-scale Deep Unsupervised Learning using Graphics Processors*, systematically discussing the feasibility of leveraging the powerful parallel computing capability of modern GPUs to accelerate deep learning training. Their work demonstrated the potential of GPUs for machine learning, particularly deep learning.
- ▶ In 2009, Google secretly initiated its autonomous vehicle project, which later evolved into Waymo. Subsequently, in 2012, vehicles developed under the project received autonomous vehicle testing licenses from the state of Nevada, becoming among the earliest autonomous vehicle programs to obtain legal testing approval.
- ▶ In 2010, the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) was officially launched and gradually evolved into one of the most influential annual competitions in computer vision. The competition witnessed numerous milestones in artificial intelligence, particularly deep learning, including the dramatic improvement achieved by the convolutional neural network AlexNet in image classification in 2012, which accelerated the widespread adoption of deep learning in computer vision.

蓬勃发展（2011~至今）

Booming (2011~Up to Now)

- ▶ 2012年，安德鲁·吴（Andrew Ng）和杰夫·迪恩（Jeff Dean）通过构建一个神经网络模型，利用谷歌的分布式计算框架，并将数以千计来自YouTube的视频输入模型进行训练，模型可以在训练后自主识别出视频中的猫。这个成就标志着深度学习技术在计算机视觉领域的突破，开创了更广阔的人工智能研究方向。
- ▶ 2012年，多伦多大学的研究人员Alex Krizhevsky等设计的一个卷积神经网络，在基于ImageNet大规模视觉识别挑战赛中取得了仅16%的错误率，相比前一年最佳成绩的25%错误率，取得了显著的提升。该网络后来被称为AlexNet。它采用了多层卷积层和池化层，使得网络能够从原始像素级别学习图像的特征。此外，AlexNet还引入了ReLU（修正线性单元）激活函数，这一非线性函数的使用有助于加速训练过程，避免了传统激活函数的一些问题。此外，AlexNet还采用了Dropout技术，可以随机地将一些神经元置为0，有助于减少过拟合。
- ▶ AlexNet在基于nVidia支持CUDA功能的消费级GPU上进行训练，成本为一般实验室所可以接受。AlexNet的成功激发了对深度学习的广泛兴趣，为后续更深层、更复杂的卷积神经网络的发展铺平了道路。它不仅对计算机视觉领域产生了深远影响，还对整个人工智能领域的发展起到了重要作用。
- ▶ 2013年，DeepMind公司开发了基于深度强化学习的模型DQN（"Deep Q-Network"）。DQN是一种深度卷积神经网络，被用来训练智能体从环境中学习，并做出优化的决策。它模型随后被用于解决多种类型的强化学习问题，特别是在视频游戏中表现出色。这些取得展示了深度强化学习在解决复杂问题方面的巨大潜力。

蓬勃发展 (2011~至今)

Booming (2011~Up to Now)

- ▶ In 2012, Andrew Ng, Jeff Dean, and their colleagues constructed a large-scale deep neural network model and trained it on massive amounts of YouTube video data using Google's distributed computing framework. Experimental results demonstrated that the model was able to autonomously learn high-level visual concepts such as “cats” without human annotations. This work demonstrated the ability of deep neural networks to automatically learn feature representations from large-scale data and is widely regarded as an important milestone in the development of deep learning, significantly advancing research in computer vision and representation learning.
- ▶ In the same year, a convolutional neural network designed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton at the University of Toronto achieved a breakthrough in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), reaching an image classification error rate of approximately 16%, significantly outperforming the previous year's best result of approximately 25%. The model later became known as AlexNet.
- ▶ AlexNet employed a deeper convolutional architecture consisting of multiple convolutional and pooling layers, enabling the network to automatically learn hierarchical image features directly from raw pixels. In addition, AlexNet introduced the Rectified Linear Unit (ReLU) activation function, which alleviated gradient vanishing issues commonly encountered in deep network training while improving training efficiency. It also adopted the Dropout technique, which randomly deactivates subsets of neurons during training to reduce the risk of overfitting.

蓬勃发展（2011~至今）

Booming (2011~Up to Now)

- ▶ Notably, AlexNet could be trained on consumer-grade GPUs supporting CUDA, making deep neural network training increasingly affordable for ordinary research laboratories. The success of AlexNet sparked widespread enthusiasm for deep learning and paved the way for the development of deeper and more sophisticated convolutional neural networks such as VGG, GoogLeNet, and ResNet. Its influence extended far beyond computer vision and profoundly shaped the broader development of artificial intelligence.
- ▶ In 2013, DeepMind developed the Deep Q-Network (DQN), a model based on deep reinforcement learning. DQN combined deep convolutional neural networks with the Q-learning framework, enabling agents to learn decision-making policies directly from environmental states (such as pixel inputs) and autonomously optimize their behavior. The model subsequently achieved remarkable performance on Atari video games, demonstrating the tremendous potential of deep reinforcement learning for solving complex decision-making problems and becoming an important milestone in reinforcement learning research.

蓬勃发展（2011~至今）

Booming (2011~Up to Now)

- ▶ 2016年3月，谷歌DeepMind的AlphaGo战胜了围棋冠军李世石。AlphaGo的胜利标志着人工智能领域取得了重大突破，展示了人工智能在高度复杂的策略性游戏中的潜力。AlphaGo结合了强化学习、深度学习和博弈理论，通过自我对弈不断提升，对于解决其他领域的复杂问题提供了积极的启发。
- ▶ 2017年，AlphaGo团队开发的AlphaZero在围棋、国际象棋和将棋等游戏中战胜了世界冠军，显示出其广泛适用于多个领域的能力。
- ▶ OpenAI于2018年、2020年分别发布了语言生成模型GPT-2、GPT-3，展示了强大的自然语言处理能力。
- ▶ 2020年，DeepMind开发的AlphaFold在蛋白质结构预测领域实现了突破，为生物医学研究提供了有力工具。
- ▶ 2021年，人工智能在自动驾驶领域取得重要进展，多家公司推出更加安全和智能的自动驾驶汽车。
- ▶ 2022年，基于GPT-3.5架构，OpenAI推出了ChatGPT，其具备强大的理解和生成自然语言文本的能力。其上下文学习能力，使得ChatGPT能够胜任回答问题、提供信息、用户会话等多种任务，引领了此后为数众多的LLM的发展。

蓬勃发展 (2011~至今)

Booming (2011~Up to Now)

- ▶ In March 2016, AlphaGo, developed by DeepMind, defeated world champion Lee Sedol in the game of Go. AlphaGo's success is widely regarded as a major milestone in the history of artificial intelligence, demonstrating AI's potential in solving highly complex strategic problems. AlphaGo combined deep neural networks, reinforcement learning, Monte Carlo Tree Search (MCTS), and ideas from game theory, while continuously improving its strategy through extensive self-play. This work not only advanced game-playing AI but also provided important insights into solving complex decision-making problems.
- ▶ In 2017, the AlphaGo team developed AlphaZero, which demonstrated a more general learning capability. Relying solely on game rules and self-play, AlphaZero achieved superhuman performance in Go, chess, and shogi, illustrating the broad applicability of reinforcement learning methods to different classes of complex decision-making problems.
- ▶ OpenAI released GPT-2 (Generative Pre-trained Transformer 2) and GPT-3 (Generative Pre-trained Transformer 3) in 2019 and 2020, respectively. These large language models demonstrated substantially enhanced natural language understanding and generation capabilities, performing strongly on tasks such as text generation, summarization, translation, and question answering. Their success also accelerated research in natural language processing based on large-scale pretraining.

蓬勃发展（2011~至今）

Booming (2011~Up to Now)

- ▶ In 2020, AlphaFold2, developed by DeepMind, achieved a major breakthrough in protein structure prediction and demonstrated near experimental-level accuracy in the Critical Assessment of Structure Prediction (CASP) competition. This achievement significantly advanced biomedical research, drug discovery, and molecular biology.
- ▶ Around 2021, artificial intelligence continued to make progress in autonomous driving. Many companies advanced the development of Advanced Driver Assistance Systems (ADAS) and autonomous driving technologies. Although fully autonomous driving still faces challenges related to safety, regulation, and adaptation to complex environments, AI has become one of the core technologies driving intelligent transportation.
- ▶ In 2022, based on the GPT-3.5 series of models, OpenAI introduced ChatGPT. With its strong natural language understanding and generation capabilities, along with its ability to perform in-context learning, ChatGPT became capable of handling tasks such as question answering, information provision, text generation, and multi-turn dialogue, while also catalyzing the rapid development and widespread adoption of subsequent Large Language Models (LLMs).